

Distributed Denial of Service (DDoS) attacks: This type of assault attempts to jam the network with distributed packets from multiple points or sources. It can be simulated using mmWave ns-3 so that the resource depletion in a 5G network that will choke the network can be predicted. Once the threshold of maximum users or traffic is analysed, a preventive algorithm can be developed for the network.

Application level attacks: This refers to the application or front-end level injection of packets to damage the network environment. In the traditional way, application level attacks are implemented by injecting malicious packets to the network channels so that confidential information can be accessed or damaged. The mmWave ns-3 module provides the libraries and code snippets for the simulation of such assaults.

The simulation of resource constrained environments is a very important task as a huge cost is associated with actual implementations. In case of cellular or wireless networks, there is need to integrate numerous resources including towers, base stations, smartphones, computing devices, servers, etc. It is very difficult to test an upcoming next-generation network using actual devices and infrastructure. For such test implementations, simulation needs to be performed so that the behaviour of a hypothetical network with a varying number of nodes can be predicted. There

are many network simulation tools and libraries that can be used to drag and drop towers, mobile handsets, smartphones, storage services, etc. Using such simulators, the behaviour of a network can be predicted. The parameters to be evaluated include security, energy, power, battery consumption, integrity, and many others. **END** 🐧

References

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etc, etc, etc?



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